

Everything You Wanted to Know About PWAs



This article is the first of a two-part series that covers all the basics of progressive Web apps (PWAs). The next part will take us further down the road to actually building a PWA.

A progressive Web app (PWA), as the name suggests, uses current Web capabilities to provide an app-like experience to users. It is a kind of application software delivered through the Web, built using common Web technologies including HTML, CSS and JavaScript. It is supposed to work on any platform that uses a standards-compliant browser. Basically, a progressive Web application takes advantage of the latest technologies to combine the best of Web and mobile apps. Imagine a website built using Web technologies but which acts and feels like an app. Recent advances in browsers, in the availability of service workers, and in the cache and push APIs have enabled Web developers to let users install Web

apps on their home screens, receive push notifications and even work offline.

The main features of PWAs

Listed below are a few key features of PWAs.

PWAs are reliable: They load immediately and never show the ‘downasaur’, even in uncertain network conditions.

PWAs are fast: They react quickly to user interactions with silky smooth animations and no junky scrolling.

PWAs are engaging: They feel like natural apps on the device, giving an immersive user experience.

Requirements of a PWA

In order to be considered a progressive Web app, it should be:

Progressive – It should work for every user, regardless of browser choice, because it is built with progressive enhancements as a core tenet.

Responsive – A PWA should work with a touchscreen interface on a smartphone, tablet or computer monitor.

Connectivity-independent – The app should be enhanced with service workers to work offline or on low quality networks.

App-like – It should use the app-shell model to provide app-style navigation and interactions.

Fresh – It must always be up-to-date, thanks to the service worker update process.

Safe – PWAs are secured by TLS encryption so your data will be secure when you’re using the app.